IoT Application for Smart Inventory Management System Based on RFID

Souhir Bousselmi
Research Laboratory Smart Electricity & ICT, LR18ES44
National Engineering School of Carthage
University of Carthage, Tunis, Tunisia
souhir.bousselmi@laposte.net

Moez Gannouni
Moveable bridge of Bizerte
Ministry of Equipement and Housing
Tunis, Tunisia
mouez.gannouni@mehat.gov.tn

Kaïs Ouni
Research Laboratory Smart Electricity & ICT, LR18ES44
National Engineering School of Carthage
University of Carthage, Tunis, Tunisia
kais.ouni@enicarthage.rnu.tn

***Abstract*— Inventory management entails calculating products or items available in stock. It represents a crucial tool for industries that face a growing demand for diverse products, where delays can cause significant problems. However, traditional inventory management models are inefficient and generate errors. In this paper, we present an IoT-based application for a smart inventory management system using Radio Frequency Identification (RFID). The RFID technology involves real-time identification of objects without physical contact, where data is stored remotely and transmitted through radio wave communication. It has become increasingly common, especially in warehouse management. The proposed innovate system aims to enhance the efficiency and precision in inventory tracking and warehouse management processes. Indeed, the integration of RFID with IoT devices enables efficient capture and analysis of real-time data at inventory levels, item locations, and stock movements. Information about products is stored on RFID tags that are attached to those items. Real-time data processing was carried out on the Raspberry Pi 4 board and visualized using a web application for supervision and control. The obtained results, using the prototype development system, demonstrate substantial improvements in inventory accuracy, reduced labor costs and optimal operational performance.**

Keywords—Inventory Management, RFID, Internet of Things (IoT), Industry 4.0

I. INTRODUCTION

The Internet of Things (IoT) is a rapidly expanding field encompassing a vast network of connected devices, which continuously generate and share data [1]. It has driven innovations across a wide range of sectors, including industry, agriculture, healthcare, science, etc [1][2]. The Industrial Internet of Things (IIoT) is a subset of IoT that signifies the fourth industrial revolution; known as Industry 4.0 [3]. It includes a wide range of applications, such as predictive maintenance, industrial automation, robotics, and inventory management [4][5]. Various organizations and companies face significant challenges in inventory and stock management. Efficiently managing warehouse storage information is crucial to the success of any company. Inventory management focuses on the strategies and actions taken by an organization or a company to determine the quantity of available products, the quantity that must be ordered, and the location of these products. The primary goal of inventory management is to strike a balance between minimizing storage costs and maintaining the ability to meet high demand. Different methods can be considered in inventory management. However, all approaches revolve around two key variables, namely the order receipt date and available quantity. In most industrial companies, inventory management primarily relies on the annual replenishment method. Every year, an order is placed based on inventory that details the available quantity of each product. The main purpose consists in restoring the stock to its maximum level and making necessary adjustments, which can lead to significant cost savings. The advantage of this method is its ability to ensure an optimal quantity of stored items while effectively managing financial investments. Exploiting the IoT technology offers enhanced visibility into inventory across stores, warehouses, and supply chains [6][7][8][9]. Based on sensors, location, status, and stock levels, each product can be tracked in real-time [10]. To ensure continuous visibility and traceability of items or products in stock, various capture technologies, such as barcodes, biometric identification, magnetic cards, and radio frequency identification (RFID) can be used [10][11]. RFID technology has become increasingly prevalent, as it not only offers an effective solution for automating stock-flow management (input flow and output flow), but also identifies and tracks scanned or tagged products [12][13][14][15]. It is the best alternative to barcodes for inventory management. Unlike barcode tags, RFID tags are more reliable and can be read multiple times without replacement. Moreover, reading barcodes can be a tedious process that often requires exact alignment between the barcode and the scanner. Other issues can occur if a barcode falls off or gets scratched, making it ineffective and unable to be scanned. A smart inventory management system combines hardware, software, and Internet of Things (IoT). However, the system's intelligence is not exclusively dependent on the IoT technology. In fact, the smart inventory system is an ecosystem, where connected hardware is complemented by a sophisticated software or an intuitive user application [6]. This system automates inventory management leading to highly accurate tracking. Furthermore, it can be enhanced with proactive and predictive alert notifications. There are various IoT-based technological solutions for intelligent inventory management [10]. Among these solutions are radio-frequency identification and sensor technologies, such as weight and ultrasonic sensors. The usefulness of weight sensors in management systems lies in their ability to precisely determine the quantity of items in stock by comparing the total weight and the weight of a single item, using a microcontroller board and a computer software. However, they can only be adopted for inventory stored on racks [16][17]. Using ultrasonic sensors, the stock level is accurately calculated by measuring the distance based on the

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propagation time of an ultrasonic wave. This solution is generally suitable for products stored on containers in a liquid form [10][18]. The main contribution of this work consists in designing and developing innovative technologies and software solutions for inventory management, based on the RFID technology and the Internet of Things. This solution helps companies whose demand is increasing for diverse items to overcome manual management models, which entail low efficiency and heavy workload. Furthermore, it allows the real-time monitoring of stock status by replacing physical inventories and sending notifications and alert messages to managers in the minimum stock event. The remainder of this paper is as follows: Section 2 reviews the RFID technology and its application for inventory management. Section 3 describes the proposed architecture of the system. The required steps for implementation and functional block diagrams are presented in Section 4. Simulation results are presented and discussed in Section 5. Finally, the present paper concludes with a summary of the main findings and insights for future research.

II. BACKGROUND

A. RFID system

Radio Frequency Identification (RFID) is a technology that aims to identify objects, track their movements, monitor their positions in real-time based on an RFID tag attached to the object that emits radio waves.

1) Operating principle

RFID focuses on the emission of an electromagnetic field by the reader. It is received by the antenna of a tag. The reader includes a coil that generates a magnetic field when powered by a voltage supply. This field activates the tag when it is close to the reader. Once activated, the tag sends a signal to the reader. This signal sets up a dialogue between the two entities, enabling data exchange. The RFID tag contains memory bytes for storing information. Figure 1 illustrates the principle of an RFID system [19].

2) RFID reader

The RFID reader is composed of a circuit that emits electromagnetic energy through an antenna, and an electronic chip that receives, decodes, and transmits the information sent by the tag to the data collection device [10].

3) RFID tag

The RFID tag is an electronic circuit that includes an antenna connected to a microchip, which receives and responds to radio requests emitted by the reader. It is responsible for storing the data used to identify the object to which it is attached. Passive tags are commonly used in warehouse management applications. However, active tags do not have an internal power source; instead, they draw energy from RFID readers [10].

B. RFID in inventory management

RFID technology is an excellent solution for stock management. It helps companies and organizations of all sizes and across various industries to enhance their inventory management processes. This technology intends to address the challenges faced by warehouses and stores in inventory management. It aims to replace an ineffective software used to ensure product traceability and reduce human errors. Furthermore, RFID enables effective automation of item storage processes and logistics flows, in alignment with systems connected to Industry 4.0 and IoT. Product information is stored in RFID tags and remotely transmitted in real-time to an application. RFID tags can be attached to any item of stock, such as product groupings, packaging cartons, pallets, plastic bins, etc., to facilitate tracking and identification. This solution is part of Auto-ID or AIDC (Automatic Identification and Data Capture) that requires neither physical nor visual contact, which is not the case for barcodes.

III. THE PROPOSED SYSTEM ARCHITECTURE

To design and develop our smart inventory management IoT solution, we have used the RFID RC522 module and the Raspberry Pi 4 board, as described below. Figure 1 illustrates the block diagram of the proposed system.

A. The description of the IoT solution

The IoT solution architecture considered in this smart inventory system includes three different layers, namely the perception layer, network layer and application layer. The perception layer is formed by a Raspberry Pi4 card and an RC522 RFID module. This module is composed of a reader and rewritable tags. At the network layer, Wi-Fi serves as a communication technique that facilitates the interaction between different layers. At the application layer, we suggest to design a web application for real-time supervision and control of stock status. For data storage, we have chosen MySQL due to its robust features. For data communication, we have used an API that includes the HTTP protocol.

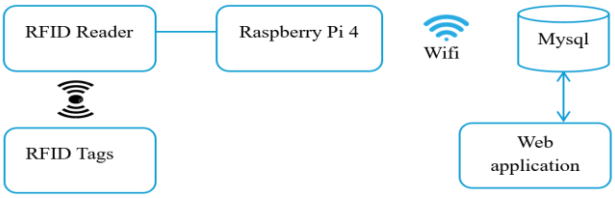


Fig 1: The block diagram of the proposed system

B. Hardware modules

1) RC522 RFID module

The RC522 module is an interface board compatible with Arduino and Raspberry Pi. It enables contactless identification using a badge or a Mifare-type RFID key [9][10]. It is based on the Phillips RC522 integrated circuit. The Raspberry Pi (or Arduino) board and RFID module communicate via the SPI bus, freeing up other ports on the card for additional applications. The communication range (distance) between RFID reader and tag is approximately 1 cm, and the operating frequency is 13.56 MHz. The choice of the reader depends on the tag frequency and the desired reading distance. RFID tags are available in three main frequency types: low-frequency, high-frequency, and ultra-high-frequency.

2) Raspberry Pi4

Raspberry Pi 4 is a compact, Linux-based nano-computer developed by the Raspberry Pi foundation to promote the computer programming education [20]. Its lightweight design, affordability, GPIO output, and modular expansion capabilities make it an excellent choice for embedded computing. Raspberry Pi Foundation recommends Raspbian, a Debian-based GNU/Linux distribution specifically optimized for the Raspberry Pi hardware.

C. Advantages of the proposed design

1) Time Savings

The proposed IoT solution significantly improves the visibility of warehouse levels and item traceability. Additionally, it simplifies and accelerates inventory management, resulting in significant time saving.

2) Reduced risk of error

Better inventory visibility helps minimize losses due to management errors. Moreover, missing items are quickly identified.

3) Proactive inventory management

Based on optimal traceability, predicting future inventory needs and managing orders accordingly become feasible. As a result, stock-outs, overstocking, and understocking are more easily avoided.

4) Compliance and Auditability

Optimal traceability ensures regulatory compliance as well as simplifies audits and inspections through precise and easily accessible records.

5) Real-time tracking

One of the key advantages of a smart inventory management system is its ability to help store managers monitor stock levels in real-time and receive alerts when the stock reaches a minimum threshold. Consequently, potential issues can be quickly identified and addressed.

IV. IMPLEMENTATION

In this section, we address the implementation of the proposed system. First, we present the hardware setup. Then, we outline the flowchart that details an item entry and exit process within the warehouse. Besides, we develop the process of acquiring data related to a product, such as its designation, location, entry date, and exit date using both Raspberry Pi and RFID module. In this section, we will describe four major stock management interfaces, namely input flow, output flow, inventory and stock movement.

A. Hardware setup

Our solution was developed based on the RFID RC522 module and the Raspberry Pi 4 that were described in the previous section. For communication, we have chosen Wi-Fi.



Fig 2: Hardware Schemata of the Solution

Figure 2 summarizes the hardware choice for the proposed solution. The RFID RC522 module communicates with Raspberry Pi 4 via the SPI (Serial Peripheral Interface) protocol. To use the module, it is essential to enable the SPI interface on the Raspberry Pi. Programming was done in Python. The MFRC522 library must be installed on the Raspberry Pi 4 board to read and write RFID tags. Writing and collecting data related to an item, such as its location and designation, are performed using the RFID reader connected to the microcontroller board. This data is processed on the Raspberry Pi to determine the incoming and outgoing flows, as well as the remaining stock quantity. The transfer of this data to the database server is handled via Python DB API (PEP 249). A web application allows the visualization of the collected data.

B. Data colletcion and database connection

For data acquisition, we establish communication between Python and MySQL, standing for the database servers used in this work. Python simplifies the interaction with the relational database system thanks to a standard API for database access. It can manage various types of databases, with a specific API designated for each database. Recently, an API known as PEP 249 was developed to ensure consistency across Python modules used for database access. This API is based on the HTTP protocol. Several Python modules can be used to communicate with MySQL, such as the MySQL Connector Python, PyMySQL, MySQLDB, MySqlClient, and OurSQL. The most recommended modules are MySQL Connector Python and PyMySQL. In our work, we choose the MySQL Connector Python since it is compatible with Python 3.

C. Software development

Establishing a supervision system is a crucial step. Any connected device must be monitored and controlled with the highest precision to ensure accurate analysis and facilitate preventive operations. In this work, a web application is used to facilitate the visualization of data collected using the IoT solution. To develop this application, we utilized the following software tools: Framework WordPress, PhpMyAdmin and Apache. Figure 5 represents the web application structure. Below, we describe their various functionalities.

1) Input flow

The "input flow" interface represents the quantity of items entering the warehouse. It displays the designation, location, entry date, and quantity of each incoming item. In this interface, we can search for an item based on its designation, entry date, or location.

2) Output flow

The "output flow" represents the quantity of items leaving the warehouse. It provides the designation, location, quantity, and release date of each outgoing item. In this interface, we can search for an item based on its designation, location, or release date.

3) Stock movement

The "stock movement" interface visualizes both incoming and outgoing item flows, specify UID, entry/exit date, location, and design each item.

4) Inventory

Inventory refers to the quantity of items available in the warehouse. The 'Inventory' interface displays the designation, location, and the remaining stock quantity. Additionally, it allows the manager to search for an item according to its designation and location.

5) Dashboard

Using the "dashboard" interface, we can analyze the statistics of incoming and outgoing items. In addition, this interface displays the last incoming and outgoing items.

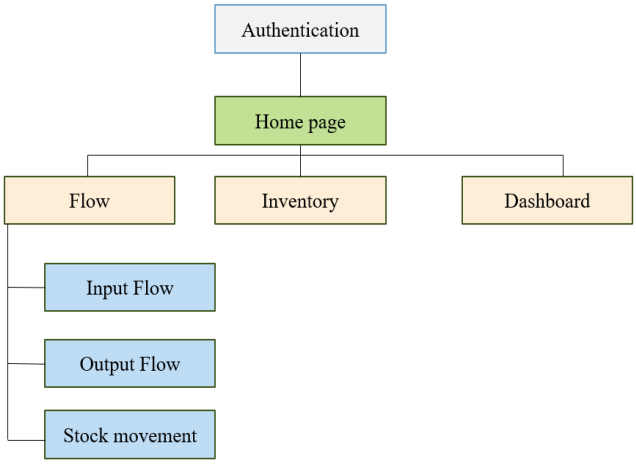


Fig 3: Flowchart of the Web Application

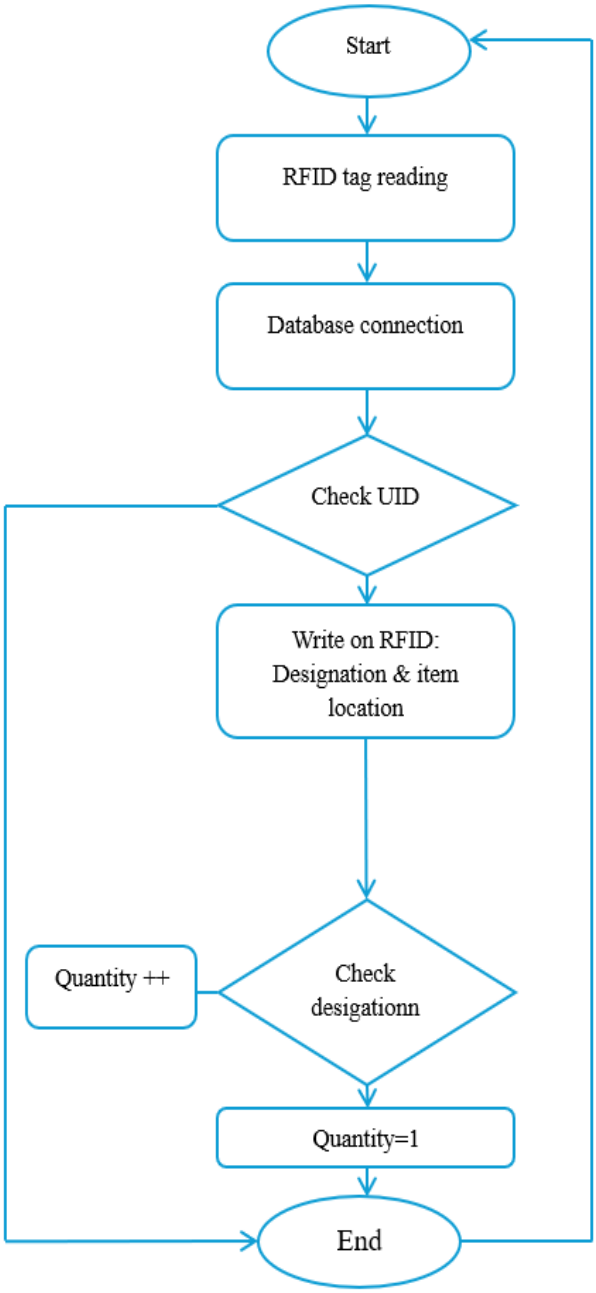


Fig 4: Flowchart of stocking new items in the warehouse

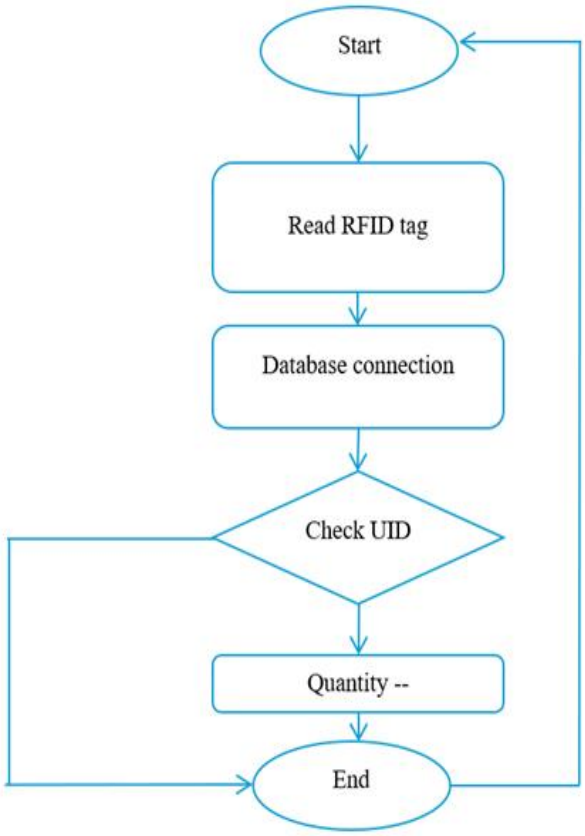


Fig 5: Flowchart of exiting items in the warehouse

D. Functional block diagrams

Figure 4 represents the flowchart that describes the process of entering items in the warehouse. The incoming items will be scanned one by one by the RFID reader. Each RFID tag represents an item with a unique UID. Subsequently, a connection to the database will be established to enter the item's UID, after checking the non-existence of the UID. If the UID is present on the database, we exit the program; otherwise, a string containing the description and location of the item is written in the RFID tag's memory. This information will be inserted into the database (e.g. the date of entry is inserted into the warehouse). For identical items (with different UIDs), quantity will be recorded in the database; otherwise, the quantity will be set to number one. Figure 5 illustrates the flowchart of the process of outcoming items in the warehouse. Outgoing items will be individually scanned by the RFID reader. The system will be then connected to the database to verify the UID. If the UID is not reported in the table, we exit the program; otherwise, the outcoming date will be inserted into the database, and quantity will be decremented.

V. SIMULATION AND RESULTS

The simulation of the developed IoT application for a smart inventory management system is conducted using a prototype store with 30 items or products. Each item corresponds to an RFID card. Hence, items with the same designation have different UIDs. Figure 6 shows a scenario of adding the item "Relay CE14" to the warehouse at location D1. The RFID card is scanned by the reader to write item data in its memory (designation, location). Figure 7 illustrates the verification of the writing process.

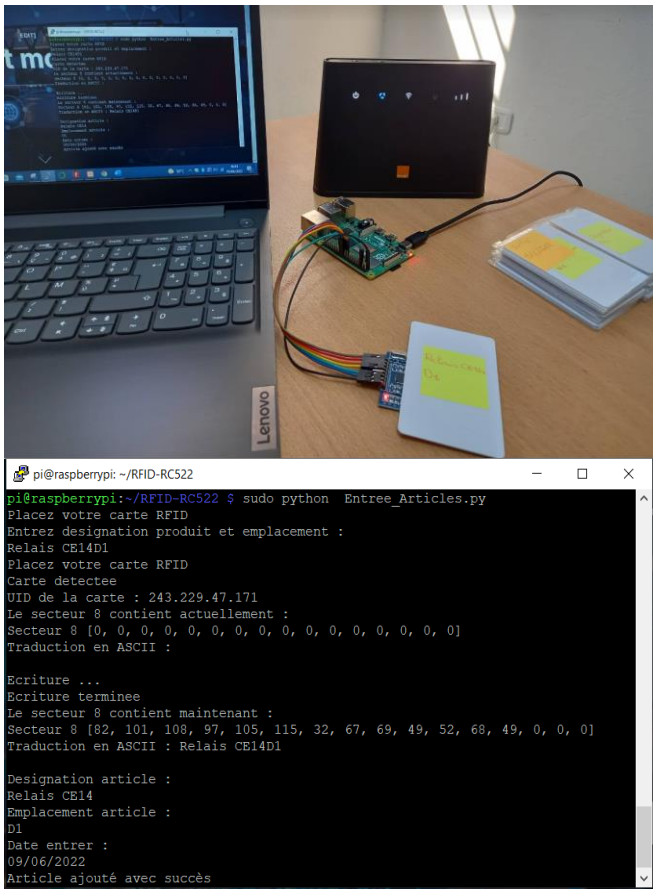


Fig 6: Scenario of entering an item into the warehouse.

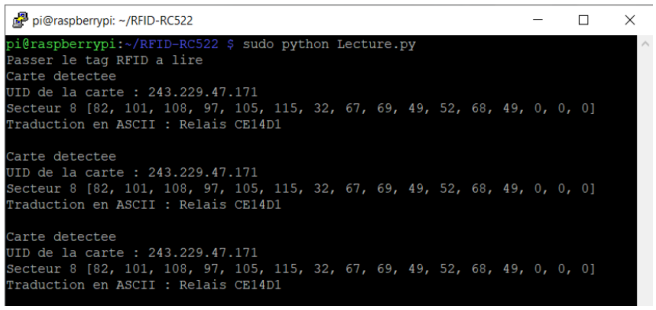


Fig 7: Check of writing an incoming item in the RFID Card

The designation, the location, and the date of entry of an item will be sent to the database. Figure 8 presents the transfer of the incoming item "Relais CE14" to the 'input flow' interface of the web application.



Fig 8 Transfer of the incoming item to the 'input flow' interface.

Figure 9 depicts a scenario of the outgoing of item "Relais CE14" (different from the entry item) from the warehouse.

Figure 10 visualizes the transfer of the outgoing date of the item "Relais CE14" to the 'output flow' interface of the web application.



Fig 9: Outgoing of an item from the warehouse



Fig 10: Transfer of the item's outgoing date to the 'output flow' interface.

The incoming and outgoing movements of the item "Relais CE14" are visualized in the 'Stock Movement' interface, as shown in Figure 11. The quantity of items remaining in stock is displayed in the 'Inventory' interface (see Figure 12) after a search based on designation.

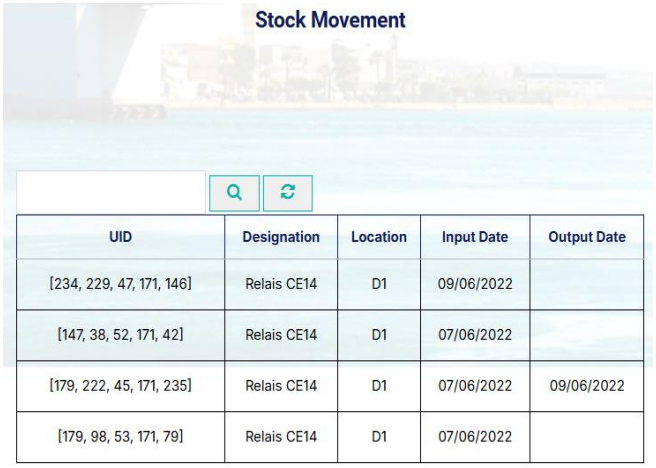


Fig 11: Visualization of the Incoming and Outgoing Movement of an Item

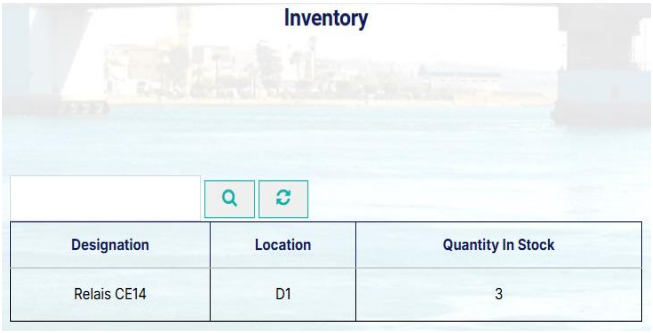


Fig 12: Displaying the Stock Quantity of “CE14 Relay” items.

We notice that this smart inventory management system is suitable for real-world applications across various sectors. Additionally, the web application can be integrated with other hardware modules, such as NodeMCU and ESP32, etc.

VI. Conclusion and Future Work

The sophisticated inventory system has directly influenced the core competencies of the enterprise. Hence, adopting smart inventory systems has become a priority. In this work, we have created a smart inventory system by using an embedded system, the Internet of Things, and a high-performance supervision application. This system generated an automated process that had a strong impact on the inventory establishment and financial management related to storage. It was based on an RFID module composed of a reader and RFID tags. Information about incoming and outgoing items collected by the RFID reader was stored on a MySQL database. Data processing was performed on the Raspberry Pi 4 board. A web application was developed to help consult both incoming and outgoing stock movements in real-time and send notifications if the stock reaches the minimum threshold. As for perspectives, we can introduce artificial intelligence to predict store needs, develop statistics, and make optimal decisions.

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